

Measurement of Flyback Converter Efficiency

Using AC Input Power Analysis

Abstract

This document describes a practical, accurate, and repeatable method for measuring the efficiency of an offline flyback power supply over a range of output loads. The technique uses time-domain measurement of instantaneous input power via an AC voltage probe and a current clamp, combined with DC output voltage measurement across a known load. The method avoids common errors associated with RMS-only measurements and is suitable for discontinuous or boundary-mode flyback converters with highly non-sinusoidal input current.

1. Purpose and Scope

The purpose of this procedure is to determine flyback converter efficiency across a defined load range using standard laboratory instrumentation. The efficiency is defined as:

$$\eta = P_{OUT} / P_{IN}$$

2. Measurement Philosophy

Flyback converters draw non-sinusoidal, phase-shifted input current due to rectification, bulk capacitance, and discontinuous magnetizing current. As a result, input power must be computed from instantaneous voltage and current rather than RMS quantities alone.

$$P_{IN} = \int v_{in}(t) \cdot i_{in}(t) dt$$

3. Required Equipment

- Digital oscilloscope with math capability
- AC-rated voltage probe (10×)
- Hall-effect current clamp (≥50 kHz bandwidth)
- Known resistive load or electronic load
- Isolation transformer
- Digital multimeter (verification)

4. Safety and Isolation

All measurements shall be performed with the DUT powered through an isolation transformer. Scope ground connections must be managed carefully to avoid inadvertent short circuits. The current clamp must be zeroed and oriented correctly prior to energizing the system.

5. Input Power Measurement

The AC input voltage is measured across line and neutral, while input current is measured by clamping around a single conductor. Instantaneous power is computed using oscilloscope math functions.

$$P(t) = V_{in}(t) \times I_{in}(t)$$

$$P_{IN} = \text{mean}[P(t)]$$

6. Output Power Measurement

For a resistive load, output current is calculated from the measured output voltage and known resistance. Output power is then computed as:

$$I_{OUT} = V_{OUT} / R_{LOAD}$$

$$P_{OUT} = V_{OUT} \times I_{OUT}$$

7. Efficiency Calculation

Efficiency is computed at each load point using the ratio of measured output power to measured input power:

$$\eta = (P_{OUT} / P_{IN}) \times 100\%$$

8. Accuracy and Common Errors

Using RMS-only measurements, insufficient averaging time, or inadequate current-clamp bandwidth can lead to significant errors. When properly executed, this method achieves $\pm 2\text{--}5\%$ absolute accuracy with excellent repeatability.

9. Conclusion

This method provides a technically correct and defensible approach for measuring flyback converter efficiency using standard laboratory equipment. By computing real input power in the time domain, systematic errors inherent in RMS-based methods are avoided.